

Project Introduction

NearHelp AI

Community Emergency Assistance System

When an emergency occurs, bystanders need quick guidance and clear location context while professional responders are on the way.

The current prototype brings an SOS entry point, a location map, emergency protocols and AI chat into one mobile experience.

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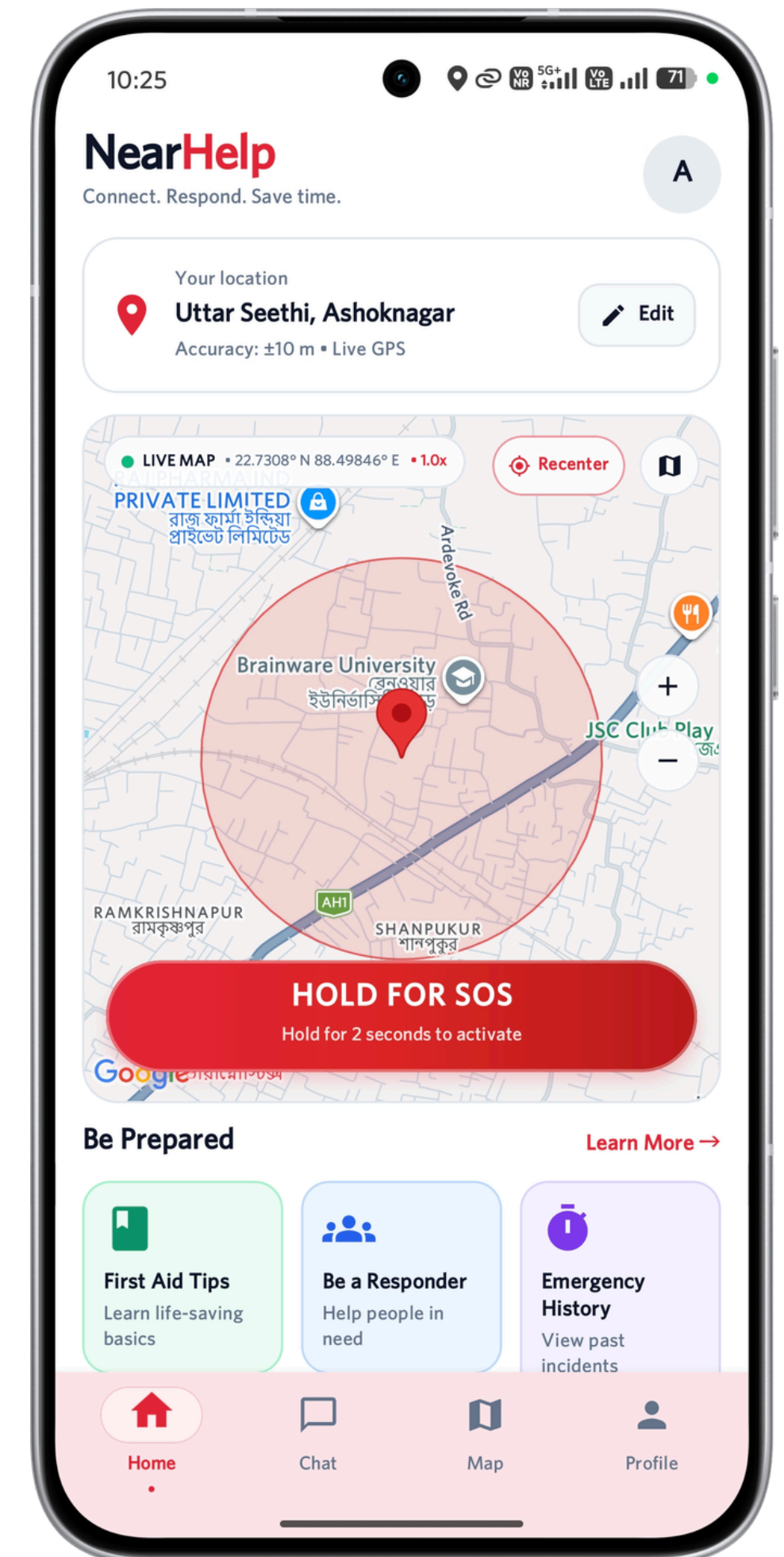
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Existing Emergency Response Challenges

The critical gap is what a person can do before organized help arrives.



Time and distance

Professional responders may be several minutes away when an incident begins.



Bystander uncertainty

People nearby may be willing to help but may not know a safe first step.



Location friction

Describing an exact location during stress can slow a request for help.



Fragmented tools

Calling, maps and guidance are often separate interactions for the user.

Project Objectives & Solution

NearHelp AI places the first emergency actions in a single Android app.



Simple Emergency Access

Provide a clear mobile interface for accessing emergency assistance quickly.



Location Awareness

Use device GPS and Google Maps to provide geographic context during an emergency.



AI-Assisted Guidance

Integrate Gemini to provide conversational emergency-related information.



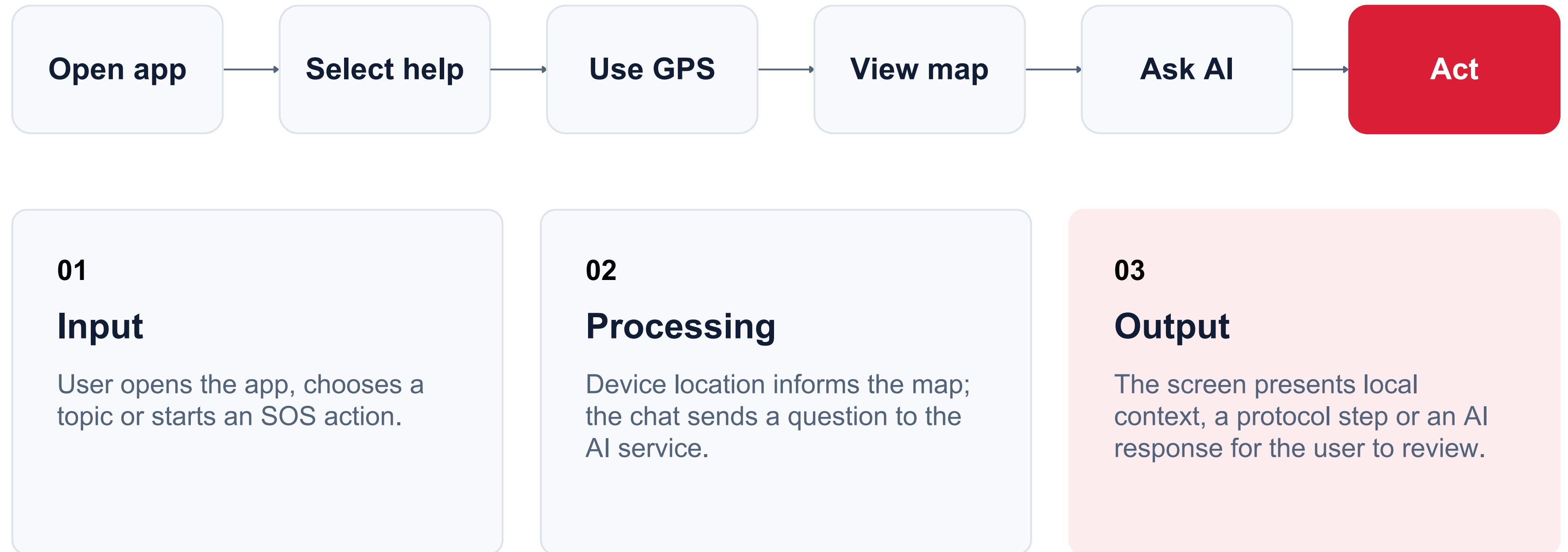
Future Community Support

Build a mobile foundation that can later connect users with verified nearby responders.

The current objective is to develop and validate the mobile foundation of NearHelp AI before implementing the complete community-response network.

System Workflow

The prototype follows a short path from opening the app to choosing an informed next action.



Implemented Features of NearHelp AI (Current Prototype)



Emergency entry

The home screen presents a hold-to-activate SOS control.



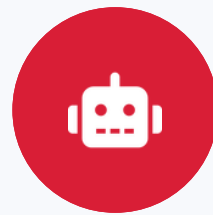
Location and map

The app shows the current GPS position and a live map view.



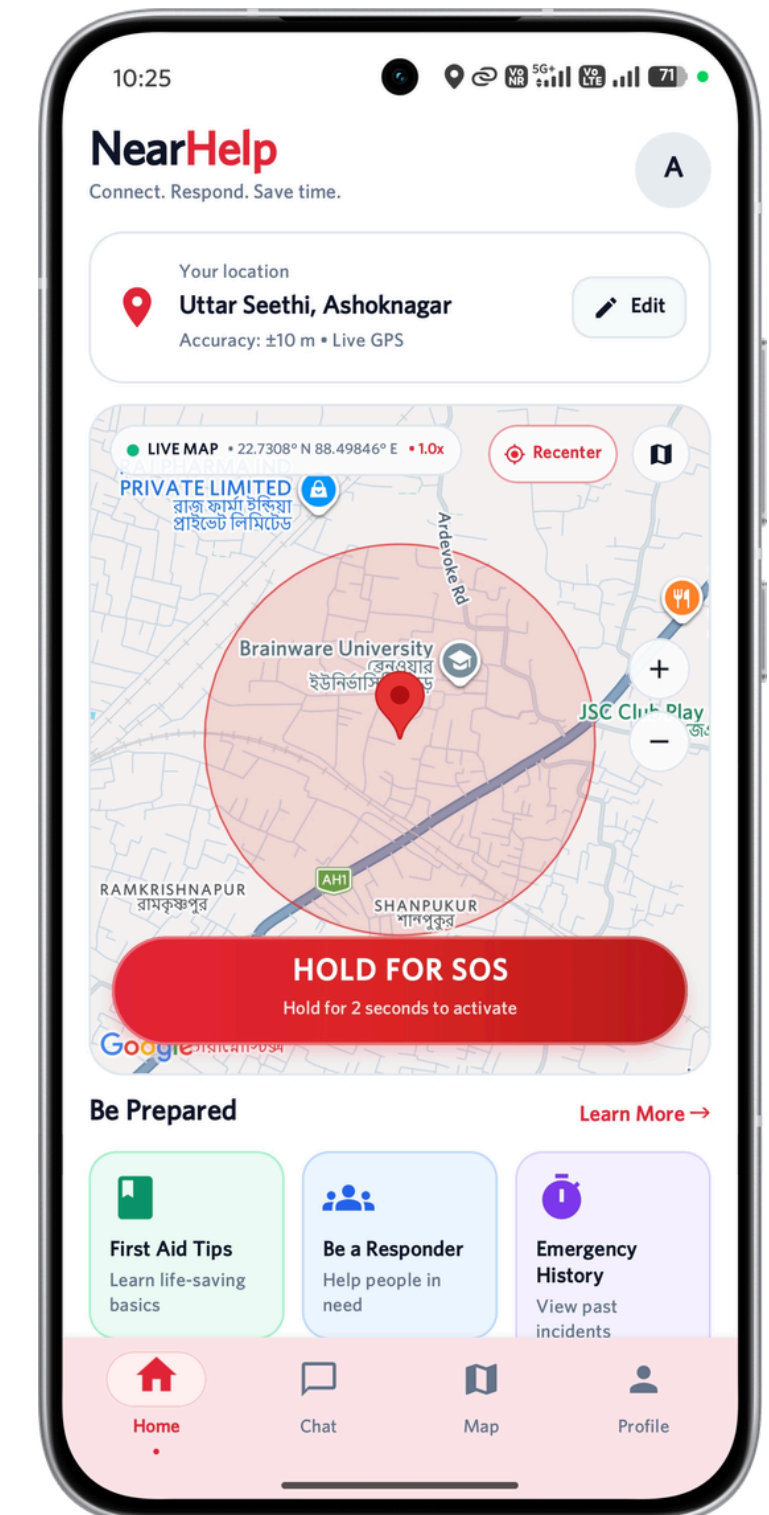
Emergency protocols

The interface displays topic-based steps for common emergencies.



AI conversation

The assistant accepts a question and displays a response.

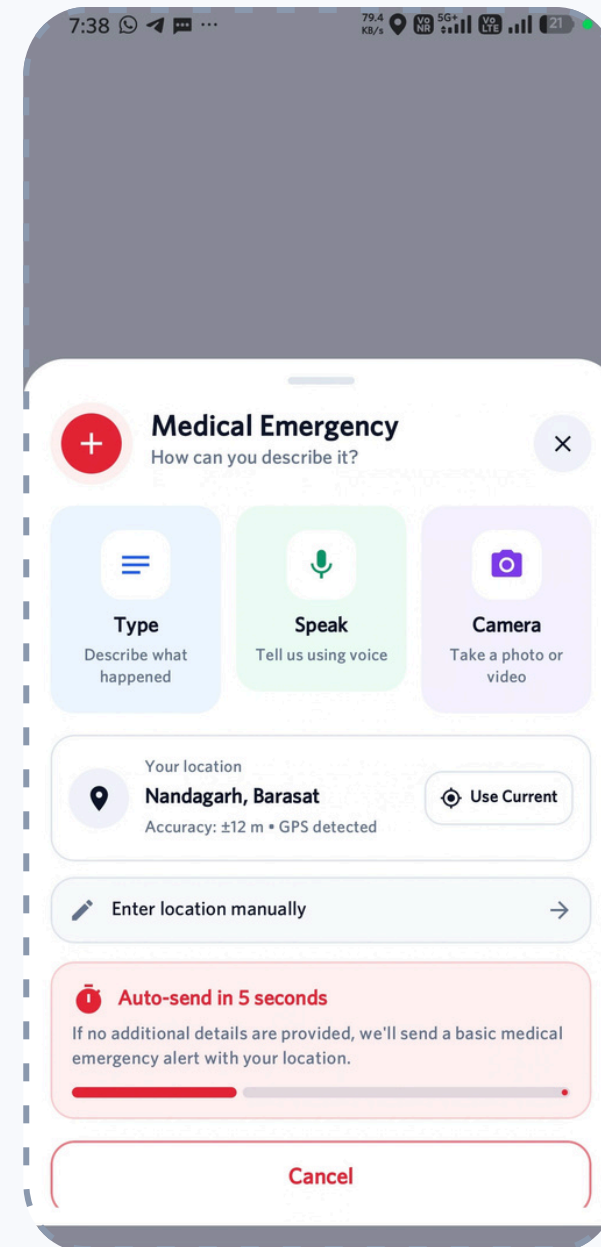


AI Chatbot & Google Maps Integration

Gemini-assisted chat

The user enters an emergency-related question. The application sends the query to the Gemini service and displays the returned response inside the chat interface.

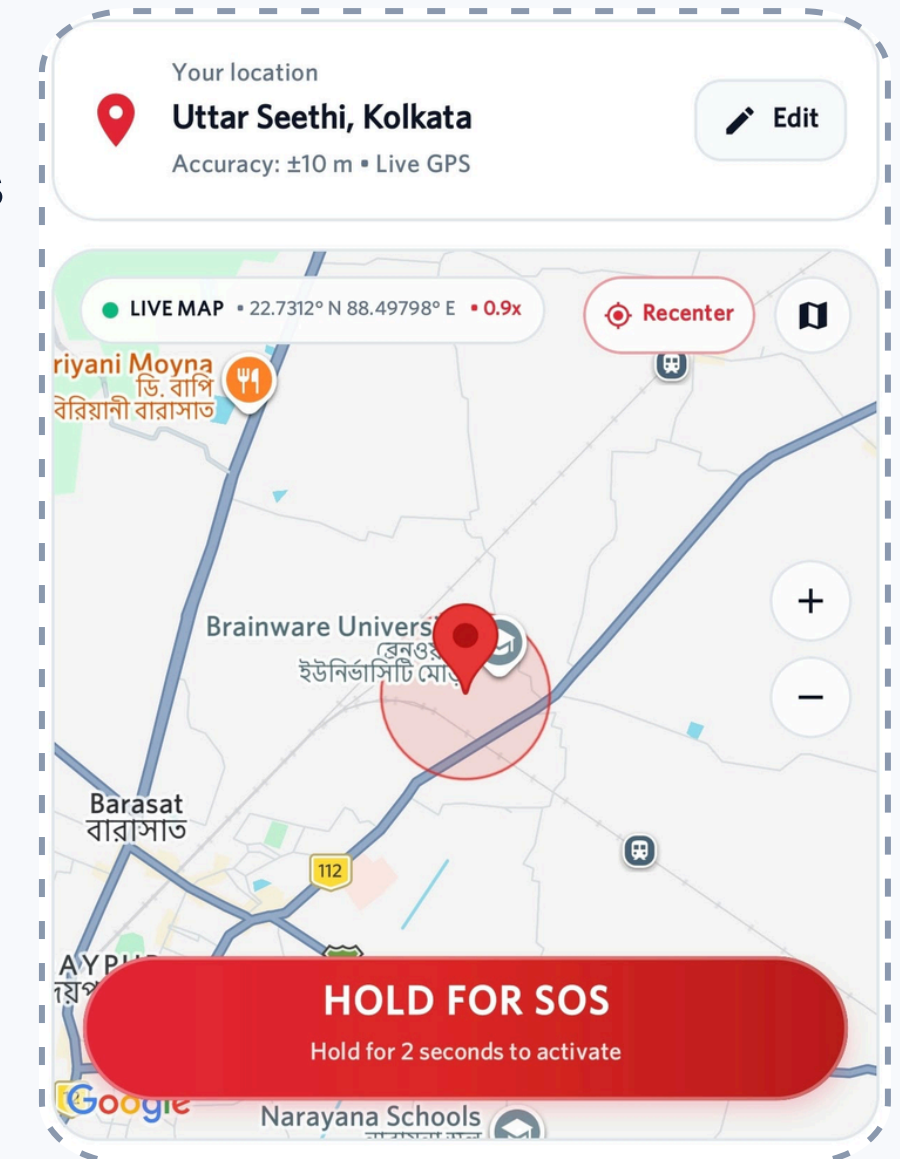
Current limitation: AI-generated emergency information requires systematic validation against verified medical sources before real-world deployment.



Location map

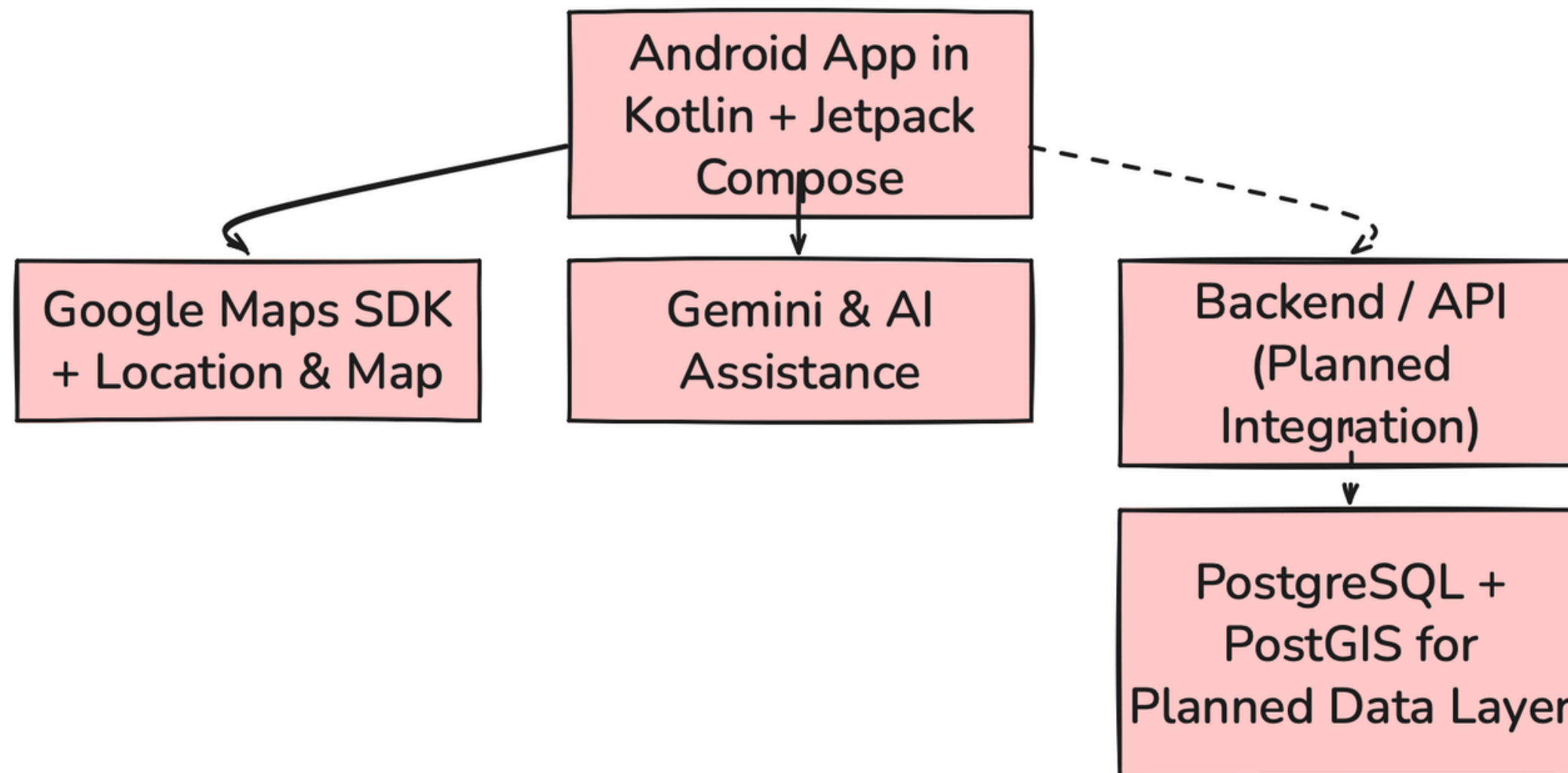
Device GPS provides the user's geographic position, while Google Maps displays the surrounding area and provides location context during emergency interaction.

The map supports orientation before an SOS action or a request for guidance.



System Architecture and Technology Stack

The Android client is the user-facing layer. It connects the location map and AI interaction.



Mobile / Frontend:

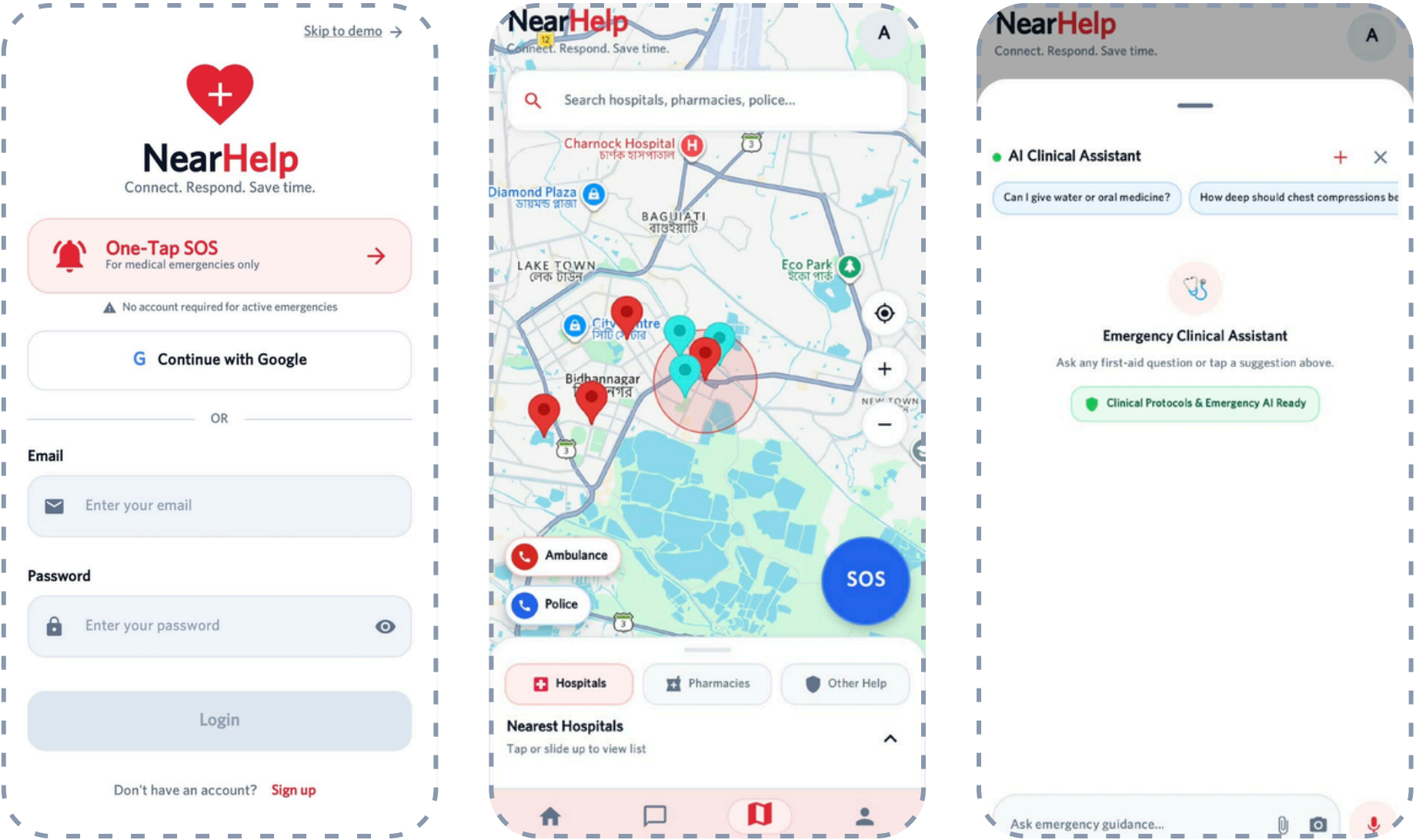
- Kotlin + Jetpack Compose — Android UI
- Google Maps SDK — Maps & location
- Gemini API — AI assistance

Backend & DB:

- FastAPI — Backend API
- PostgreSQL + PostGIS — Database & geospatial data
- Firebase / FCM — Auth & notifications
- WebSockets — Real-time coordination

Application Demo, Testing & Current Results

Current evidence comes from device screenshots of the working Android interface.



Demo check	Observed result
Launch + navigation	Home and major tabs appear
Location map	GPS area and SOS control shown
Emergency content	Protocol topic and steps shown
AI conversation	Question and response captured

These checks confirm visible interface behavior. Dispatch reliability, response time and medical accuracy need separate evaluation.

Research & Reference:

- Existing research supports the use of mobile and location-based systems to connect nearby community responders with emergency cases.
- Predictive dispatch studies show that responder selection can be improved using availability, response behaviour and location.
- AI/ML research demonstrates the potential of intelligent emergency triage and prioritization, while healthcare RAG research supports using trusted medical knowledge to ground AI-generated guidance.
- NearHelp AI integrates these approaches into one platform for AI triage, intelligent responder matching, GPS-based coordination and reliable first-aid assistance.

RESEARCH REFERENCES

01. Predictive Dispatch of Volunteer First Responders: Algorithm Development and Validation — JMIR, 2023

Research Paper — JMIR

02. Smartphone-based Dispatch of Community First Responders to Out-of-Hospital Cardiac Arrest — 2021

Supporting Research — Springer Nature

03. Application of Artificial Intelligence in Triage in Emergencies and Disasters: A Systematic Review — BMC Public Health, 2024

AI/ML Research — Springer Nature

04. Retrieval Augmented Generation for Large Language Models in Healthcare: A Systematic Review — PLOS Digital Health, 2025

RAG Research — PLOS Digital Health

Limitations, Future Scope & Conclusion

Current limitations

- AI responses require validation against verified emergency and medical protocols
- The prototype does not yet establish a complete real-world responder dispatch network.
- Real-world response time, responder availability and large-scale performance have not yet been evaluated.

Future scope

- RAG-based verified emergency knowledge
- Nearby responder matching
- Real-time rescue tracking
- Bengali, Hindi and English support
- Voice/image emergency input
- Professional emergency-service integration

Conclusion: NearHelp AI now demonstrates the mobile foundation for location-aware emergency assistance.

*Thank
you!*

ANY QUESTIONS?